

BIOGRAPHICAL SKETCH

NAME: Hannah W. Mercier

eRA COMMONS USER NAME: hwmercier

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Boston University, Boston, MA	BS	05/2007	Therapeutic Studies
University of New England, Portland, ME	MS	05/2011	Occupational Therapy
MGH Institute of Health Professions, Charlestown, MA	PhD	05/2016	Rehabilitation Sciences
Harvard Medical School, Boston, MA	Postdoctoral Fellow	05/2020	Physical Medicine & Rehabilitation

A. Personal Statement

I am an Assistant Professor of Occupational Therapy, and my research interests focus on enhancing health and participation outcomes for adults with neurological conditions. My clinical background and postdoctoral research in rehabilitation sciences created a strong interest in effects of neurological conditions and empowering self-management approaches to mitigate them, specifically in the form of mobile health. I have collaborated with colleagues in computer science, biomedical engineering, and psychiatry to develop these interventions, and have trained and supervised four graduate computer science and mathematics students to specifically refine the mobile health technology and analytical approaches to maximize clinical utility. The proposed project unites my research interests and skills to design and validate an impactful, responsive, and personalized technology enhanced intervention to engage and support the driving rehabilitation needs of adults with disabilities. Our team has developed this work through regular bimonthly meetings, and each member brings exceptional complementary skills and experience. I am uniquely qualified to serve as the Principal Investigator and lead the smartphone platform refinement, testing, and deployment, and study the use of mobile technology for engaging driving rehabilitation participants. It is my hope that this work will enhance the assessment of underlying driving skills with a more comprehensive and reliable reflection of performance and provide an accessible and effective path for supporting driving participation among adults with disabilities.

Recently awarded funding in the area of mobile health assessment within driving training includes:

- Association of Schools Advancing Health Professions Interprofessional Innovation Grant
Principal Investigator, 9/1/25-5/31/26
Community-engaged co-design of a mobile intervention tool for enhancing driving skills in neurodiverse young adults

Recently completed projects that focus on the application of mobile health assessment and interventions among individuals with disabilities include:

- National Institute on Disability, Independent Living, and Rehabilitation Research, Administration for Community Living
Switzer Research Fellowship
Principal Investigator, Award #90SFGE0024 9/1/20-8/31/23
Characterizing Patterns of Depression, Community Mobility, and Social Engagement in Sub-acute Spinal Cord Injury
- Craig H. Neilsen Foundation, Psychosocial Research Grant, Postdoctoral Fellowship
Principal Investigator, Award #542007 4/30/18-4/29/20
Digital Phenotyping to Evaluate Psychosocial Well-being and the Effect of Exercise in Spinal Cord Injury

B. Positions, Scientific Appointments, and Honors**Positions and Scientific Appointments**

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|------------|--|
| 2009-2011 | Graduate Research Assistant, Returning Warrior Project: Qualitative analysis of participation and stress among military service members upon return from combat, University of New England, Portland, ME |
| 2011- 2018 | Clinical Specialist, Occupational Therapy, Spaulding Rehabilitation Network, MA |

2012 -	Research Consultant and Co-Investigator, Empower Spinal Cord Injury Inc.
2012 - 2016	Doctoral Research Fellow, New England Regional SCI Center at Boston University Health and Disability Research Institute, MA
2013 - 2016	Doctoral Research Fellow, Spaulding Harvard SCI Model System, MA
2014 - 2018	Lecturer & Clinical Lab Instructor, MGH IHP, Occupational Therapy Doctoral Program, MA
2015 - 2020	Project Director, Spaulding Rehabilitation Network, Cardiovascular Research Laboratory, MA
2016 - 2020	Post-doctoral Research Fellow, Department of Physical Medicine and Rehabilitation, Harvard Medical School, MA
2020 -	Assistant Professor, Department of Occupational Therapy, School of Health Professions at Stony Brook University, NY

Honors

2012	Partners in Excellence Team Award, Libya Project, Partners Healthcare
2011	University of New England Student Facilitator, Interdisciplinary Symposium: Veterans Health and Welfare

C. Contributions to Science

1. Doctoral Career: My research training has provided me with a strongly laid foundation in research with human subjects and in exploiting large databases to explain the psychosocial components to activity and participation among people with disabilities. As a doctoral fellow with Dr. Alan M. Jette at the New England Regional SCI Center (NERSCIC), I took advantage of the wealth of the SCI Model Systems Database to better define the relationship of hospitalization with subsequent outcomes of disability and psychosocial function. I received the University of Alabama Birmingham Devivo Mentored Research award to complete this work, and I provided new evidence identifying characteristics of those at greatest risk for poor outcomes following hospitalization. I presented findings at several conferences, and authored publications that composed part of my doctoral dissertation. At NERSCIC I led another project to evaluate the efficacy of a responsive telehealth intervention to decrease the severity of depression among adults with SCI. I found that those who were depressed initially demonstrated significant improvements in depressive symptoms and tended to have fewer hospitalizations after the treatment. This retrospective analysis also revealed that, in comparison to a population with similar functional limitations, those with SCI were less engaged in the intervention and in turn required more extensive follow-up and individualized support. My final actions at NERSCIC were to develop a peer-led telehealth intervention applying an empowerment model to more effectively engage those with SCI to successfully self-manage their health.

- Houlihan BV, Everhart-Skeels S, Gutnick D, Pernigotti D, Zazula J, Brody M, Burnett S, **Mercier H**, Hasiotis S, Green C, Seetharama S, Belliveau T, Rosenblum D, Jette A. Empowering adults with chronic spinal cord injury to prevent secondary conditions. *Arch Phys Med Rehabil*. 2016; 97(10):1687-1695.e5
- **Mercier HW**. (June 2019). Disability after hospitalization: Impact of personal and health condition factors for persons living with spinal cord injury. Spinal Cord Injury Model Systems Project Directors meeting, Washington, DC
- **Mercier HW**, Everhart-Skeels S, Houlihan B, Zazula J, Jette A. Health empowerment approach to promote self-management and healthcare utilization in a peer-led telehealth intervention for adults with SCI. Poster presented at International Spinal Cord Society and American Spinal Injury Association Joint Scientific Meeting, 2015 May, Montreal, QC.
- **Mercier HW**, Ni P, Houlihan BV, Jette AM. Differential impact and use of a telehealth intervention by persons with MS or SCI. *Am J Phys Med Rehabil*. 2015 Nov;94(11):987-99.

2. Postdoctoral Career: I accepted a Harvard Medical School postdoctoral position under the mentorship of Dr. J. Andrew Taylor and Dr. J.P. Onnela because of the wealth of exceptional opportunities and resources available to me. In turn, I brought my background in psychosocial health and the impact of disabilities to Dr. Taylor's Cardiovascular Research Laboratory and Dr. Onnela's Digital Phenotyping Laboratory. At the Spaulding Rehabilitation Network Cardiovascular Research Laboratory, I managed measurement and analysis of psychosocial outcomes in all studies related to adapted rowing, including Dr. Taylor's R01, "Hybrid-FES Exercise to Prevent Cardiovascular Declines in Acute Spinal Cord Injury" (HL117037). With Dr. Taylor's mentorship I gained expertise implementing exercise interventions to support health and more distal participation outcomes. Additionally, I have co-authored a book chapter published in "The Physiology of Exercise in Spinal Cord Injury".

I collaborated with Dr. Onnela and psychiatrist Dr. John Torous and received their mentorship to use powerful, finer-grain data analytics to answer my behavioral research questions. My postdoctoral fellowship represented a natural integration of my research interests in the potential impact of exercise interventions on psychosocial health among people with SCI and utilizing dynamic technology and powerful statistical methods to explain within and across-individual changes. I received a Craig H. Neilsen Foundation Psychosocial Research Grant (Post-Doctoral Award) to use smartphone-based digital phenotyping for monitoring and predicting depression, anxiety, pain and community participation among those with SCI as

they underwent an exercise intervention. I also received a Spaulding Research Institute Accelerator Grant to study psychosocial wellbeing as those with new injury transition to community-living.

- **Mercier HW**, Hamner JW, Torous J, Onnela JP, Taylor JA. Digital Phenotyping to Quantify Psychosocial Wellbeing Trajectories after Spinal Cord Injury [published online ahead of print, 2020 Jun 22]. *Am J Phys Med Rehabil*. 2020;10.1097/PHM.0000000000001506.
- **Mercier HW**, Picard G, Taylor JA, Vivodtzev I. Gains in aerobic capacity with whole-body functional electrical stimulation row training and generalization to arms-only exercise after spinal cord injury. *Spinal Cord*. 2021;59(1):74-81. doi:10.1038/s41393-020-0527-2.
- **Mercier H**, Solinsky R, Taylor JA. Relationship of cardiometabolic disease risk factors with age and spinal cord injury duration. *J Spinal Cor Med*. 2022; PMID: 35485952 DOI: 10.1080/10790268.2022.2065410.
- Solinsky R, **Mercier H**, Picard G, Taylor JA. Cardiometabolic Effects of High-Intensity Hybrid Functional Electrical Stimulation Exercise after Spinal Cord Injury. *PM R*. 2021;13(9):937-944. doi:10.1002/pmrj.12507
- **Mercier HW**, Taylor JA. The physiology of exercise in spinal cord injury: An overview of the limitations and adaptations. In *The Physiology of Exercise in Spinal Cord Injury*. 2016 (pp. 1-11). Springer US.

3. Independent Investigation: I received NIDILRR funding to build on my previous findings that explored the individual-level digital phenotype of depression, community mobility, and social engagement, to apply these methods with newly-injured individuals with SCI in a post-COVID-19 landscape. The interactions between depression, community mobility, and social engagement are complex and little is known about their time-dependent nature during the first few years of community participation after SCI. This work also examined the impact of minimally intrusive smartphone-based feedback and mindfulness intervention on psychosocial wellbeing in adults with new SCI. I presented a short course on this work at a national conference. My work recognizes that newly-injured individuals with SCI have decreased clinical and peer support upon inpatient rehabilitation discharge, resulting in many demonstrating overt psychosocial issues during community life despite none during their inpatient stay. I believe this will have great impact on policy and development of feasible, effective community-based supports post-SCI. Furthermore, my present independent investigation continues to foster interprofessional collaborations that can best serve those with neurological conditions in the community. I have collaborated with Dr. Michelle Ballan to investigate intimate partner violence against women with SCI with a Craig H. Neilsen Foundation Creating Opportunity & Independence, Community Support Grant. We will use a train-the-trainer model to equip rehabilitation clinicians in their role to create a culture of empowerment and support women to prevent violence. I have continued to serve as research consultant with Empower SCI., a unique short-term residential rehabilitation program that creates a "second chance" to address their psychosocial wellbeing and independence goals, and received the inaugural Stony Brook School of Health Professions Interprofessional Seed Grant to pursue this work. I have collaborated internationally to explore psychosocial wellbeing and mobile health, presenting to the Universidad Autónoma de Madrid School of Medicine seminar and co-authoring a manuscript that covered this topic relative to the COVID-19 pandemic. Finally, I consulted with the Boston Harvard Burn Injury Model System in their move to adopt smartphone-based assessment of health and participation to understand experiences post burn injury.

Our School of Health Professions Seed Grant presents an opportunity to build on my previous work in mobile health technology, assessment, and intervention and continue to promote health and independence among adults with disabilities.

- Park JK, Ren B, **Mercier H**. (2023). Predicting pain to support exercise participation after spinal cord injury. *International Journal of Exercise Science: Conference Proceedings*; 15(3): Article 2. Available at: <https://digitalcommons.wku.edu/ijesab/vol15/iss3/2>
- **Mercier HW**, Fairman, A. (March 2022). Exploring the power and potential of smartphone technologies for people with disabilities. Short Course presented at the AOTA INSPIRE Meeting; San Antonio, TX.
- **Mercier HW**. (June 2021). Digital phenotyping methods and intervention for participation and mental health after spinal cord injury. Invited lecture at the Harvard Medical School Neurorehabilitation Conference, Waltham, MA
- **Mercier HW**. (May 2023). Smartphone-based health and wellbeing assessment: Optimizing a personalized approach after major injury. Invited lecture at the Universidad Autónoma de Madrid School of Medicine seminar.
- Castelletti C, del Castellar BD, Miret M, **Mercier H**, Lara E. (2024). Loneliness and subjective wellbeing during the COVID-19 pandemic: A systematic review. *The Journal of Psychology: Interdisciplinary and Applied*. Doi:10.1080/00223980.2024.2319172
- Deng H, Abouzeid CA, Shepler LJ, Slavin MD, Taylor JA, **Mercier HW**, Herrera-Escobar JP, Kazis LE, Ryan CM, Schneider J. Using digital phenotyping to characterize psychosocial trajectories for people with burn injury. *Burns: journal of the International Society for Burn Injuries*. 2022; <https://doi.org/10.1016/j.burns.2022.04.003>.
- Deng H, Vu K, Franco J, Shepler LJ, Abouzeid CA, Hamner JW, **Mercier HW**, Taylor JA, Kazis LE, Slavin MD, Ryan CM, Schneider JC. (2023). Digital interventions for social participation in adults with long-term physical conditions: A systematic review. *J Med Syst*; 47(1):26. doi:10.1007/s10916-023-01914-7.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: Casanova, Rosalia

eRA COMMONS USER NAME (credential, e.g., agency login):

POSITION TITLE: Clinical Assistant Professor

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Suffolk County Community College, Brentwood, NY	OTH	05/2007	Occupational Therapy Assistant
Stony Brook University, Stony Brook, NY	MOTH	06/2011	Occupational Therapy
Stony Brook University, Stony Brook, NY	BS	06/2011	Health Sciences
Boston University, Boston, Massachusetts	DOTH	09/2019	Occupational Therapy

A. Personal Statement

I am a licensed occupational therapist, certified Driver Rehabilitation Professional (DRP), and New York State certified driving instructor, with advanced clinical training in adult physical rehabilitation. I am currently the only DRP on Long Island with a healthcare background who provides both comprehensive clinical and behind-the-wheel evaluations. My professional expertise enables me to bridge the gap between medical evaluation, driver instruction, and rehabilitation research. This SHP SEED grant supports the next phase of a driver rehabilitation initiative that builds upon a prior pilot study. The initial pilot focuses on designing a mobile health application informed by Michon's model of driving behavior, with emphasis on observable traffic-level driving behaviors. This process also included mentoring an undergraduate research student who was selected and awarded a \$5,000 scholarship for research studies to collaborate on this pilot study with the PI team. In this next phase, we aim to evaluate how data from the app—gathered by passenger observers (e.g., caregivers or family members)—correlates with outcomes from formal clinical driving evaluations and self-performance ratings. The goal is to create an innovative resource tool for drivers caregivers, providers, and driver rehabilitation professionals to better understand and monitor an individual's driving performance. This project is innovative in that it introduces a mobile, community-based dyadic model where observers contribute real-time behavioral data, supporting a more holistic approach to driver safety and clinical decision-making. I bring critical expertise in evaluating medical fitness to drive, including interpreting motor, visual, and cognitive functioning. I will serve as the clinical lead in this study—overseeing participant evaluations, supervising app deployment, and analyzing relationships between app, clinical, and self-assessment data. My prior work in geriatric fall prevention, culturally responsive clinical education, and simulation-based DEI training reflects a broader commitment to advancing equitable, accessible, and evidence-based practices across rehabilitation contexts.

B. Positions, Scientific Appointments and Honors**Positions and Scientific Appointments**

2022 -	ClinicalAssistantProfessor,Stony Brook University
2020 - 2023	Supervising Occupational Therapist, The Hampton Center for Rehabilitation and Nursing, Southampton, NY
2019 - 2022	Academic Fieldwork Coordinator, Hofstra University, Hempstead , NY
2019 - 2022	Clinical Assistant Professor, Hofstra University, Hempstead, NY

2019 - 2020	Interim Director of Rehabilitation Services, The Hampton Center for Rehabilitation and Nursing, Southampton, NY
2017 - 2019	Assistant Director of Rehabilitation Services, The Hampton Center for Rehabilitation and Nursing, Southampton, NY
2011 - 2017	Occupational Therapist, The Hampton Center for Rehabilitation and Nursing, Southampton, NY
2008 - 2009	Gurwin Jewish Nursing and Rehabilitation Center , HealthPRO, Commack, NY
2008 - 2009	Occupational Therapy Assistant, Good Samaritan Nursing and Rehabilitation Center, Sayville, NY
2008 - 2009	Occupational Therapy Assistant, Out East Therapies- Home Care services, Southampton, NY
2007 - 2011	Occupational Therapy Assistant, The Hampton Center for Rehabilitation and Nursing, Southampton, NY

Honors

2023	Hispanic Heritage Month Madrina Award, Stony Brook University
2019	American Occupational Therapy Association Service Commendation, American Occupational Therapy Association

C. Contribution to Science

1. I contributed to the development of a mobile health app informed by Michon's driving behavior model. This app enables caregiver and passenger-based assessments of older adults' driving safety. This work supports a new pilot study pending August 2025 examining correlations between perceptions of driving and driving behaviors that are monitored using a mobile health app. Mercier, H., Sukhraj, S., Casanova, R. (2025–present). Feasibility of driver/passenger dyad assessment [Unpublished research study, Stony Brook University]. IRB2025-00299.
2. This work emphasizes the role of simulation in preparing future clinicians for disaster response and real-world crisis decision-making. I designed and developed simulation protocols, roles, and integrating the responses into graduate level occupational therapy curriculum to address the role of occupational therapy practitioners during emergency responses that a therapist may face in the clinical setting. I developed and presented a poster highlighting the use of simulation to teach occupational therapy students emergency response skills. Casanova, R., & Coffin, D (2024, March 21). Enhancing entry-level education: embedding emergency response activities through simulation [Poster session]. American Occupational Therapy Association Inspire Annual Conference, Orlando, FL, United States
3. As co-investigator on a mixed-methods study exploring quality of life among amputee soccer athletes, I contributed to the IRB-approved study design and collaborated closely with the research team to develop demographic questions, study methods, questionnaires, and the interview protocol. I conducted all participant interviews and supported the team in analyzing themes from athlete responses. This research highlights the significant impact of adaptive sports—specifically amputee soccer—on participants' perceived quality of life and psychosocial well-being. Lee, D, Pierre-Glaude, J, Lamberg, E, Casanova, R.(2023–Present). Amputee Soccer Interviews. IRB2023-00489, Stony Brook University.
4. As co-investigator on two funded projects, I helped design and deliver simulation-based DEI training focused on microaggressions in healthcare. One project developed an interactive workshop for clinical educators; the second employed simulated scenarios to train students in responding to microaggressions in the clinical environment. These initiatives address systemic issues in clinical education, improve equity in supervision, and empower both educators and students to create safer, more inclusive care environments. Komnik, L, Pierre-Glaude, J, Casanova, R.(2025). Tools for Addressing Microaggressions: An Interactive Workshop for Health Professions' Clinical Educators. Funded by ASHP (\$2,000), AWD00004419. Pierre-Glaude, J., Komnik, L., Casanova, R.(2024).

Breaking Barriers in the Clinical Setting: Using Simulation for Microaggression and Bias Training. SBU Presidential Mini Grant (\$3,520).